Actual Hermite loss and source-precision stability

November 9, 2026. Mathematical source `a375671ad`.

Unsigned producer evidence throughout [pcc:unverified].

Result and domain. The actual normalized Hermite multiplier has a quantified valuation bound. The actual local reciprocal and evaluation loss are source-connected. An explicit finite recovery functional preserves precision using the source's original child precision. The exact reconstruction identity remains separate. The error theorems here concern the concrete rational integral ring $\mathcal{O} = \{r \in \mathbb{Q} : |r|_2 \le 1\}$. Coordinate refinement to the extension/nilpotent lattice and complete bit costs remain open. No curve or subgroup solve, benchmark or novelty claim is made.

Actual basis and inverse. Let $L = 2^a$, $\alpha = 2L$, $t_m = m - L$ on the actual grid, $h = \lceil 3M/u \rceil$, $q(0) = 0$ and $q(t) = h \max(0, u - v_2(t))$ otherwise. Construct

$$E_t(T) = \prod_{s \neq t} \left(\frac{T - s}{t - s} \right)^{q(s)}, \qquad E_t(t + \alpha z) = \prod_{s \neq t} \left(1 + \frac{\alpha z}{t - s} \right)^{q(s)}.$$

The zero-node factor has exponent zero. Each actual nonzero difference has absolute value below $2L$; the established ratio theorem puts $\alpha/(t - s)$ in $\mathcal{O}$. The actual local product has constant one, so the proof constructs its integral reciprocal and its finite mapped inverse buffer. It does not assume a unit for an unspecified basis.

Actual multiplier bound by pairing. Exclude $0, t$ and let $m(s)$ be the actual centered representative of $t - s$. Lean proves the involution, both exclusions, congruence modulo $2L$ and exact valuation preservation. Put $v_s = v_2(s)$, $w_s = v_2(t - s) = v_2(m(s))$, $i = v_2(t)$. For the actual product at zero,

$$V = v_2(E_t(0)) = \sum_s q(s)(v_s - w_s), \qquad 2V = \sum_s (q(s) - q(m(s)))(v_s - w_s).$$

The decreasing natural taper is h-Lipschitz, giving $2V \ge -h \sum_s (v_s - w_s)^2$. The actual ultrametric relation gives

$$(v_s - w_s)^2 = (v_s - i)_+^2 + (w_s - i)_+^2.$$

Pairing makes these two sums equal. Both excluded terms have zero excess. Actual level counts, including the lone endpoint at top level a, imply

$$\sum_s (v_s - i)_+^2 = \sum_{j=0}^{a} 2^{a-j} (j - i)_+^2 \le 6\, 2^{a-i}.$$

The final bound uses the finite identity $\sum_{k=0}^{n} k^2/2^k = 6 - (n^2 + 4n + 6)/2^n$. Thus $V \ge -6hD_i$, $D_i = 2^{a-i}$, from the actual grid rather than an assumed moment estimate or random distribution.

Evaluation loss and actual coefficient errors. The actual ceiling obeys $hu \le 4M$ for $u \le M$. For $b, t \in \mathcal{O}$, $b \neq 0$, an explicit integral numerator proves

$$b^q P(t/b) = \sum_{j<q} P_j t^j b^{q-j} \quad (\deg P < q).$$

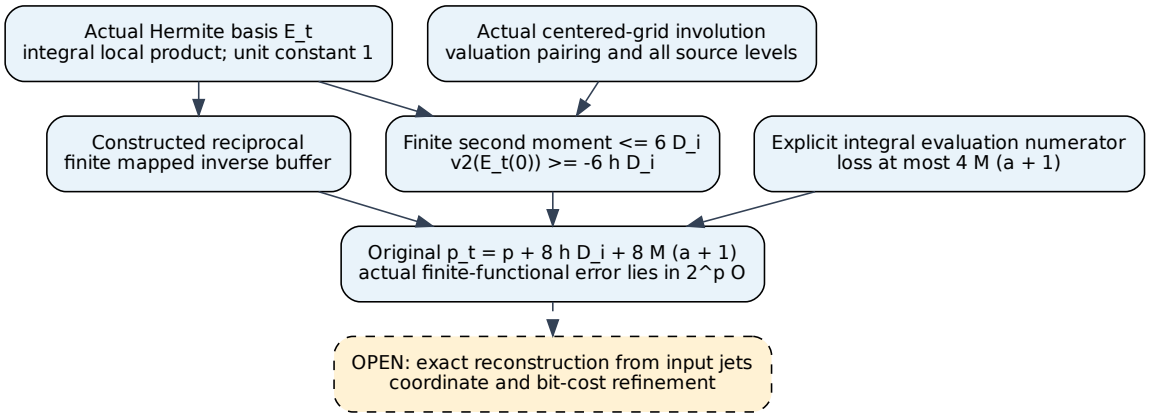
With $b = 2L$, $q = 4M$, this bounds the actual evaluation loss by $K = 4M(a+1)$. Let $B = 6hD_i$. The source's original child precision splits as

$$p_t = p + 8hD_i + 8M(a+1) = p + B + K + S, \quad S = 2hD_i + 4M(a+1) \ge 0.$$

A coefficient error $2^{p_t}c$ contributes, after division by 2^p,

$$2^S \left(2^B E_t(0)\right) \left(2^K c(-t/\alpha)^j\right) \in \mathcal{O}.$$

Summing coefficients and actual sampled nodes proves that the explicit `recoveryFunctional` on supplied bounded quotient polynomials preserves precision p. Multiplication by the actual local reciprocal preserves input finite buffers. No two-power is cancelled inside a finite ring.



Solid boxes are this component; the dashed reconstruction bridge remains open.

Remaining exact bridge. Construct exact quotient truncations from an input polynomial, prove the D.9 interpolation polynomial's jets and degree, and apply existing uniqueness to obtain its value at zero. The stability theorem does not assume or prove that reconstruction equality. Full native refinement and recursive costs remain separate. Earlier affine/source-recovery reports remain immutable additions.

Proof map and unsigned PCC records. Canonical statement hashes are in `lean/hermite-precision-claim-pins.json`.

Actual integral chart	`HermiteBasis.actualBasisLocalEval` PCC-20261109-09b188

Finite mapped reciprocal	`HermiteBasis.`
	`actualBasisMappedInverseBuffer`
	PCC-20261109-348f1c

Actual mirror valuations	`HermiteGrid.actualMirrorValuation` PCC-20261109-f0b5d3
Actual multiplier loss	`HermiteMoment.actualHermiteMultiplierLoss` PCC-20261109-827aa4
Actual evaluation loss	`HermiteBasis.actualEvaluationLoss` PCC-20261109-ad5a31
Finite-functional stability	`HermiteError.actualRecoveryFunctionalStable` PCC-20261109-00cb3f

Input-buffer precision	`HermiteError.`
	`actualLocalQuotientBufferStable`
	PCC-20261109-fd958d

Checks and pins. Default build passed. The actual source-discovering repository gate reports 216 declarations and zero extra trusted axioms. Foundations remain `propext`, `Classical.choice`, `Quot.sound`. Fresh import-inclusive replay exited0; see `lean/hermite-precision-fresh-status.json` and its raw log. All checks are unsigned producer evidence [pcc:unverified].

Pinned Lean4.35.0, CryptoLean `148f5dc389b341401e06d76abe7783c4ccf42f2e`, Mathlib `cfc55fc58714a59dc06e6e79edc6d1d95140d3c2`. Failed build/API diagnostics are retained. No checker, gate, trust policy or dependency revision changed. Quantified finite sums, valuations and buffer statements are covered by Lean, rather than polynomial-expression certificate v1.

Sources and scope. Read the pinned OpenAI October25 manuscript D.9–D.10 and the prior matched-recovery review, checking actual Mathlib APIs. Primary PDF SHA256: `5b1612dc2eebb2cad5b4cbe62d1fed2982f01ab1908c26f70010a651487e8123`. Provenance: `research/prime_common_kernel_20261030/sources.md`. KB unavailable (no MCP, wrapper exit127). This is bounded formalization, not an exhaustive novelty audit.

Next comparison. Only after exact reconstruction and cost refinements should a separately prospective native campaign compare the actual original precision against the old doubled-budget proxy. This proof provides a mathematical reason to revisit that proxy without changing a historical panel number. No old run was repeated, no benchmark row changed and no outcome was attributed to CPU contention. The versioned diagram records this component's proof dependencies.